Problem Set #9 – Fundamental Algorithm Techniques

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Problem 1 — Finite Functions

Consider a finite Boolean domain:

$$F : \{0,1\}^n \to A$$

1. For output $A = \{0,1\}$ there are 2^{2^n} possible functions.

   Explanation: The domain $\{0,1\}^n$ has 2^n different inputs. For each input, we choose one of 2 possible outputs. Therefore,

   $$|\text{Functions}| = 2^{2^n}.$$

2. For output $A = \{-1,0,1\}$ there are 3^{2^n} possible functions:

   $$|\text{Functions}| = 3^{2^n}.$$

3. For output $A = \{0,1\}^m$, each output has 2^m choices. Thus,

   $$|\text{Functions}| = (2^m)^{2^n} = 2^{m \cdot 2^n}.$$

Problem 2 — NAND Generates AND, OR, NOT

The NAND operation is defined as:

$$A \uparrow B = \neg(A \wedge B)$$

1. **NOT gate**:

   $$\neg A = A \uparrow A$$

2. **AND gate**:

   $$A \wedge B = (A \uparrow B) \uparrow (A \uparrow B)$$

3. **OR gate**: Using De Morgan:

   $$A \vee B = \neg(\neg A \wedge \neg B)$$

   Therefore,

   $$A \vee B = (A \uparrow A) \uparrow (B \uparrow B)$$

Thus, NAND is a universal gate.

Problem 3 — Universality of Boolean Circuits

For any function

$$F : \{0,1\}^n \to \{0,1\},$$

we define a delta function for each input x:

$$\delta_x(y) = \begin{cases} 1, & y = x, \\ 0, & y \neq x. \end{cases}$$

Each δ_x can be implemented as:

$$\delta_x(y) = \bigwedge_{i=1}^{n} \begin{cases} y_i, & x_i = 1, \\ \neg y_i, & x_i = 0, \end{cases}$$

which requires $O(n)$ gates.

Then the function F can be written as:

$$F(y) = \bigvee_{x:F(x)=1} \delta_x(y)$$

There are at most 2^n such x, therefore total complexity:

$$O(n \cdot 2^n).$$